

What does a language model detect when it detects an injected thought?

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Abstract

Recent work reports that large language models can detect when a concept vector has been injected into their residual stream, and reads this as evidence of introspective access. The published false-positive control is the absence of injection. We ask whether the model's report reflects the content of what was injected or only the fact of a perturbation. To separate the two, we compare real concept vectors against two vectors of the same magnitude that carry no concept: a Gaussian direction rescaled to the real norm, and the real vector with its coordinates shuffled. On Gemma3-27B-it, at the layer and prompt of the original work, we first reproduce the reported detection rate (10.0% against 10.8%, with zero false positives). We then measure the probability that the model's first generated token is "yes", so that no output yet exists for it to read. At the pre-registered operating point, a content-free vector produces the whole effect (real 0.417, random 0.305, $p=0.33$). At lower strengths a concept-specific component is present but accounts for a minority of the shift, and it appears just as strongly under a prompt that never mentions the model, its mind, or injection. The self-directed prompt also raises the response to content-free vectors eightfold relative to the neutral prompt. In this setting, a report of detecting an injected thought is mostly a report of having been perturbed. The residual concept-specific signal does not depend on asking the model about itself, and the no-injection control cannot distinguish either finding from introspective access. Every criterion was filed before the run it governs. One of our own hypotheses failed under that discipline and is reported.

1. Introduction

Can a language model tell when its own activations have been altered? Macar, Yang, Wang, Wallich, Ameisen and Lindsey (2026) inject a concept vector into Gemma3-27B-it at layer 37 and ask the model whether it detects an injected thought. The concept vector is the residual-stream difference between thinking about a concept and a baseline. They report a 10.8% detection rate with a 0% false-positive rate, trace the behaviour to a circuit that appears only after post-training, and treat the result as a defended case of introspective access. Later critiques (Lederman and Mahowald, 2026; *Reality Check*, 2026) attack the identification half of the claim, whether the model can say what was injected, and leave detection standing.

The detection claim rests on a contrast between injection and no injection. That contrast cannot separate three things the model might be doing. It might be reporting access to the injected content. It might be reporting that something was done to its activations, whatever that something was. Or it might simply answer "yes" more often because its computation was disrupted, with no report involved at all. A model that has been kicked behaves differently from one that has not, and a question inviting it to say so will get a "yes". None of that requires knowing what the kick contained.

The control that separates the first possibility from the other two is an injection with no content: a vector of the same magnitude that carries no concept. That control had not been run. We run it.

We also test whether detection depends on asking the model about itself. If a prompt that never mentions the model, its mind, or injection produces the same response, then the word "introspective" in "introspective access" is doing no work.

Our contributions are the following.

- A reproduction of the reported detection effect on free hardware, in 4-bit, with the original prompt, layer, and scoring rule.
- A demonstration that the detection rate reported from generated text is a property of the readout. The same injection yields 50% or 7% depending only on the strength chosen, while the first-token probability is flat.
- The content-free control, pre-registered. At the operating point matching the published number, a random vector reproduces the effect. At lower strengths a concept-specific residual exists but is a minority of the shift.
- The framing control, pre-registered. A prompt that never refers to the model is indistinguishable from the introspective prompt once anything is injected, and is less susceptible to content-free perturbation.
- A pre-registered hypothesis of our own, that the generated-text detections were the model inferring from its own steered output, which the first-token measurement refuted.

We take no position on whether these models can introspect. We report what the standard protocol measures and what it does not.

2. Method

Model and hardware. We use google/gemma-3-27b-it, the primary model of the original work, loaded in 4-bit NF4 with float32 compute across two Tesla T4 GPUs on Kaggle. Float16 compute is not viable for this model on this hardware. The residual stream at layer 37 reaches a peak coordinate magnitude of 51,436 against a float16 ceiling of 65,504, and some prompts overflow. Because every concept vector shares one baseline mean, a single overflow poisons all of them silently.

Concept vectors. We follow the original Appendix B.2. For each concept c , we take the residual-stream activation after decoder block 37 at the last token of the chat-templated prompt "Tell me about c ", and subtract the mean of the same activation over 30 unrelated common nouns. There are 30 concepts, 20 concrete and 10 abstract. The median vector norm is 5,002. We do not normalise the vectors; the original does not state its convention.

Injection. We set h to $h + \alpha \cdot v$ at layer 37 on prompt positions only. Where generation occurs, it proceeds without injection.

Prompts. The introspective prompt is the original text. A preamble states that the researcher can inject patterns into the model's activations and does so on about half of trials, followed by "Trial n : Do you detect an injected thought? If so, what is the injected thought about? Answer starting with YES or NO." The neutral prompt keeps the structure and the stated base rate but never mentions the model, its mind, its activations, or injection: "I am a researcher running a series of trials. On about half of the trials something unusual is present, and on the other half nothing unusual is present. Trial n : Is anything unusual present? If so, what is it about? Answer starting with YES or NO."

Two readouts. The generated-text readout uses greedy decoding for 60 tokens, scored as in the original: a detection requires an affirmative answer and a coherent response. We take the affirmation to be whichever of YES or NO the model emits first. The first-token readout is the probability assigned to YES over YES-or-NO at the first generated position, computed from logits in a single forward pass, with all casing and leading-space variants of each token pooled. Nothing is generated, so nothing exists for the model to read back.

Content-free controls. The random control is a Gaussian vector rescaled to the norm of the real vector for that concept. The shuffle control is the real vector with its coordinates permuted. Both preserve magnitude, and the second also preserves the marginal distribution of coordinate values. Both are seeded and reproducible.

Analysis. Every comparison is paired by concept and tested with a two-sided Wilcoxon signed-rank test at 0.05. Pairing also holds constant a design flaw we discovered after the first run and discuss in Section 5: concept identity was perfectly confounded with the trial number in the prompt.

Pre-registration. We wrote and committed the criterion for each stage before that stage ran, including what we would conclude under each outcome. We fixed the primary test for the content-free control at $\alpha=6$, the strength at which our generated-text detection rate matched the published one. One hypothesis of ours was refuted at the first pre-registered stage and is reported in Section 3.3.

3. Results

3.1 The effect reproduces

Under the generated-text readout with the introspective prompt at $\alpha=6$, 3 of 30 trials were scored as detections. That is 10.0%, Wilson 95% CI 3.5% to 25.6%, against the published 10.8%. At $\alpha=0$ the count was 0 of 30, matching the published 0% false-positive rate. The experiments that follow concern an effect that is present in our setup.

3.2 The reported rate is a property of the readout

The same model, layer, prompt and vectors give very different numbers depending on how the answer is read.

α	detection, generated-text readout	P(YES), first-token readout
0	0%	0.000
2	50%	0.429
4	23%	0.504
6	27%	0.417
8	7%	0.430

The first-token probability is flat from $\alpha=2$ upward. The generated-text rate ranges from 7% to 50% over the same interval. Reading the 210 generated responses shows why. At high strength the model's output is dominated by the injected concept and never reaches an answer. At low strength it answers readily, and the answer is about nothing in particular. A detection rate from generated text is therefore a joint property of

the model, the strength, and the decoding budget, and it is not comparable across papers that differ in any of them.

3.3 The first-token shift is real, and our hypothesis about it was wrong

We predicted that the generated-text detections were the model reading its own output. In those trials, every scored detection had the concept-laden text before the affirmative answer. One response read, in full: "betrayal, betrayal, betrayal, loyalty, loyalty, loyalty, betrayal... YES. The repeated words suggest the concept of 'betrayal' and 'loyalty' was injected." We pre-registered the hypothesis that this was the mechanism, and predicted that the first-token probability, where no output exists to read, would not move.

It moved. Under the introspective prompt, $P(\text{YES})$ at the first token was 0.00003 at $\alpha=0$ and 0.417 at $\alpha=6$, with $W=3.0$, $p=9.3 \times 10^{-9}$, rising on 28 of 30 concepts. Whatever the model is responding to, it responds before generating anything. The hypothesis is refuted, and the pre-registered outcome labelled A is the one we report.

3.4 Most of the shift is a response to being perturbed

The pre-registered primary test compares real concept vectors against norm-matched random vectors under the introspective prompt at $\alpha=6$.

vector	mean $P(\text{YES})$
real concept	0.417
random, norm-matched	0.305

The difference is not significant ($p=0.33$), with real higher on 17 of 30 concepts. This is the outcome we had labelled A2: at this strength the model responds to the magnitude of a perturbation and not to its content.

The declared sweep across strengths shows where content matters and where it stops mattering.

α	real	random	shuffle	real vs random	real vs shuffle
2	0.429	0.184	0.128	$p=0.0011$	$p=0.0099$
4	0.504	0.252	0.359	$p=0.0040$	$p=0.13$
6	0.417	0.305	0.374	$p=0.33$	$p=0.76$

A content-free vector alone lifts $P(\text{YES})$ from 0.00003 to between 0.13 and 0.37 at every strength, with $p < 10^{-8}$ on 28 to 30 of 30 concepts. On top of that generic response sits a concept-specific one, significant at $\alpha=2$ and $\alpha=4$ and absent at $\alpha=6$. The share of the real effect that a content-free vector reproduces is 36% at $\alpha=2$, 61% at $\alpha=4$, and 82% at $\alpha=6$. The two controls agree with each other at every strength ($p=0.50, 0.27, 0.60$).

We placed the primary test at $\alpha=6$ because that is where our generated-text rate matched the published figure. It is also where the generic response has saturated and the concept-specific one is least visible. Matching a published number selected against the signal we were looking for. We report the sweep because it was declared, and we report the $\alpha=6$ result as primary because it was pre-registered as such.

3.5 Asking the model about itself adds nothing

The pre-registered secondary test compares the introspective and neutral prompts with real vectors.

α	introspective	neutral	p
0	0.000	0.188	<0.0001
2	0.429	0.370	0.44
4	0.504	0.481	0.75
6	0.417	0.395	0.70

The prompts differ only when nothing is injected. Once something is injected they are indistinguishable. The concept-specific component also survives the change of framing: under the neutral prompt, real vectors beat random at $\alpha=2$ and $\alpha=4$ ($p=0.0087$ for both). Whatever the model tracks, it tracks it equally well when it is not asked about itself.

3.6 The self-directed prompt inflates the generic response

The two prompts respond very differently to a content-free vector at low strength. The rise in $P(\text{YES})$ over the no-injection baseline for a random vector at $\alpha=2$ is +0.184 under the introspective prompt and +0.023 under the neutral prompt. Under the neutral prompt a real vector at the same strength rises +0.182, so the neutral prompt separates content from mere perturbation while the introspective prompt does not. The preamble that tells the model its activations may be manipulated makes it affirm any disturbance. As an instrument for concept detection the neutral prompt is cleaner. It has a higher floor, but the floor is not driven by content-free perturbation.

3.7 What the detections are made of

Reading the generated responses shows what an affirmative answer contains. Of the 15 affirmative answers at $\alpha=2$, none named the injected concept. Seven described the experiment back to us, with phrases such as "a researcher studying my activations" and "a feeling of being observed and evaluated". Three said "red apple", which was never injected, and two more said so at $\alpha=4$. When the injected concept did surface in a response it was overwhelmingly the correct one, at 145 times the rate at which a non-injected concept surfaced ($p=1.7 \times 10^{-7}$ at $\alpha=6$), but it surfaced where the model no longer affirmed a detection. Affirmation without content occurs at low strength, content without affirmation at high strength, and the two coincided on three trials.

4. Discussion

A zero false-positive rate against no injection establishes that the model does not affirm a detection spontaneously. It does not establish what the model is affirming when it does. Our data say that at the published operating point the affirmation is fully reproduced by a vector with no content, and that at lower strengths it is mostly so. On this evidence, the claim "the model detects an injected thought" is mostly "the model detects an injection".

A concept-specific component remains and we do not dismiss it. At $\alpha=2$ the real vector beats both content-free controls under both prompts. The component is small, it fades as the generic response grows, and it does not depend on self-directed questioning. It is compatible with the model having some access to the

content of a perturbation. It is also compatible with real concept vectors being less disruptive than random ones of the same norm, since an on-manifold direction may perturb the computation differently from an off-manifold one. We cannot separate those two explanations here.

The neutral prompt reaches the same response as the introspective one at every injected strength. If the mechanism involved a representation of the self, a question that does not invoke the self should engage it less, and it does not. The one thing the introspective framing changes is the response to content-free perturbation, which it raises eightfold at low strength. A prompt that primes the model to expect interference is not a neutral probe of whether it notices interference.

The generated-text detection rate is the number the literature reports, and Section 3.2 shows it swinging from 7% to 50% across strengths over which the first-token probability does not move. The published figure is therefore a statement about a decoding procedure as much as about a model. Any comparison across papers, models, or strengths that does not fix the readout is not a comparison.

We do not show that the model cannot introspect, nor that the circuit reported in the original work does not exist. We show that the behavioural protocol on which the detection claim rests admits a non-introspective explanation that its own controls cannot exclude, and that the control which can exclude it costs one forward pass per trial.

5. Limitations

We tested one model, one layer, and one quantisation, with 30 concepts against the original 500 and rule-based scoring against the original's language-model judge. The generated-text figures in particular should be read as reproductions of a pattern rather than of a number.

Trial number was confounded with concept. In every run reported here, concept i was always "Trial $i+1$ ", and the trial number appears in the prompt. Pairing by concept holds it constant in every test above, but it is visible in the neutral prompt's no-injection baseline, where several concepts in the 20s show P(YES) above 0.6 with nothing injected. The experiment script now assigns trial numbers by seeded shuffle and records them. The runs here predate that fix.

Our α scale is not the original's. We sweep and report every strength, but the strength at which the published rate was matched is not necessarily the strength the original used.

The first-token readout pools YES and NO over casing and whitespace variants. Other answer forms such as "Absolutely" or "I do not detect" are not captured by it, though the generated-text readout does capture them.

We tested one neutral prompt. A prompt-sensitivity sweep would strengthen Section 3.5.

6. Reproducibility

All runs execute on a free Kaggle session with two T4 GPUs in under 30 minutes each after model load. The experiment script, the analysis scripts, raw per-trial data for every condition, and the pre-registration documents with their filing dates are in the project repository. The script prints a version stamp as its first line of output, and every table above can be regenerated from the archived JSONL files without GPU access.

References

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[Wilson score interval and Wilcoxon signed-rank test: standard references to be added.]